Lab 12 — Cables and Connectors — Identification and Selection

> **Course:** CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)
> **Topic 03:** Hardware — Core 1, 25% of Core 1
> **Exam objective:** Identify every cable and connector on the A+ objective list by sight and select the correct one for a stated requirement (Core 1 objectives 3.1 and 3.2).

Goal

You identify cables and connectors from physical samples or high-resolution references, record the specification that limits each one, and then solve a set of selection scenarios where choosing the wrong cable would fail. The output is a selection guide organised by the question it answers.

What you'll produce

A completed cable and connector identification table and a solved set of eight selection scenarios.

Tools and equipment

Cable and connector samples or reference images, specification sheets, measuring tape

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Build the identification table with columns for connector name, cable type, what it carries, maximum speed, maximum distance and typical use.
2. Identify the network group: RJ45 for Ethernet, RJ11 for telephone, F-type for coaxial, LC, SC and ST for fibre, and record what distinguishes each visually.
3. Record the twisted-pair categories with their speed and distance limits: Cat 5e at 1 Gbps to 100

m, Cat 6 at 10 Gbps to 55 m, Cat 6a at 10 Gbps to the full 100 m.

4. Identify the video group: HDMI, DisplayPort, DVI and VGA, and record for each whether it carries digital or analog video and whether it carries audio.
5. Identify the peripheral group: USB-A, USB-B, USB-C, micro-USB, mini-USB, Lightning and Thunderbolt, and record the speed each generation supports.
6. Identify the storage group: SATA data and power, eSATA, M.2, and the legacy IDE/PATA 40-pin and SCSI 80-pin ribbon connectors.
7. Solve scenario one: a 120-metre run between two buildings needs 1 Gbps. Copper cannot exceed 100 m, so fibre is the only correct answer — record your justification.
8. Solve scenario two: a cable run passes beside fluorescent lighting and a motor room. Record why shielded twisted pair or fibre is required and unshielded is not.
9. Solve scenario three: a user needs to connect a modern laptop with only USB-C to an old VGA projector. Record why an active adapter is required, since VGA is analog and USB-C is digital.
10. Solve scenario four: a 4K display at 120 Hz is needed. Record which connector versions support this and which do not.
11. Solve the remaining scenarios covering an external drive at maximum speed, a multi-monitor daisy chain, a legacy serial console connection and a PoE camera run.
12. Complete the selection guide organised by the question asked: how far, how fast, digital or analog, and does it carry power.

Test it — verification

Your identification table covers at least 20 connectors with speed and distance limits; every scenario answer names a specific cable or connector and justifies it against a stated limit rather than a preference.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the `apt-get install` step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against `labs/tools.md`. All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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